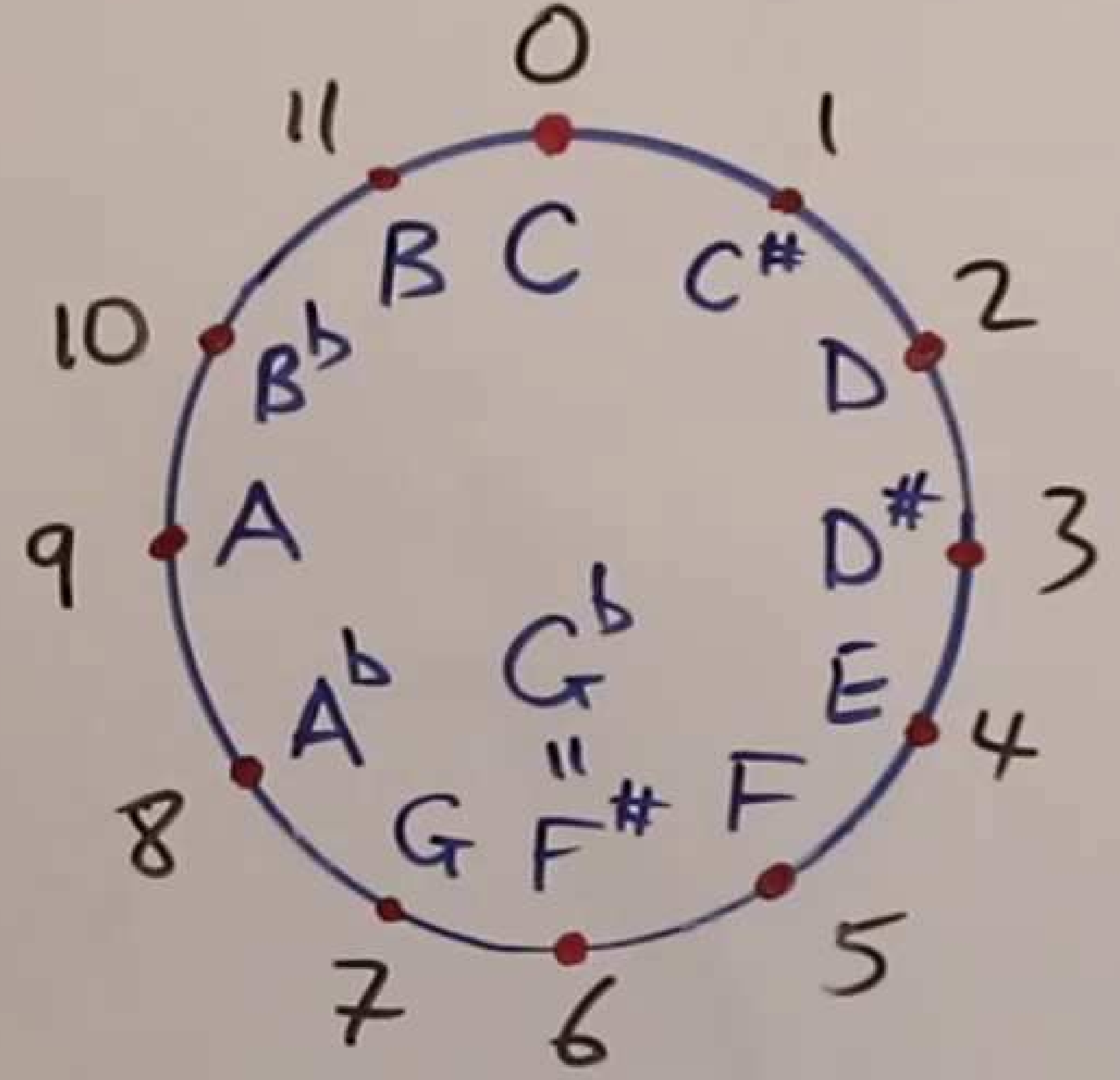
The impossibilities of a Pythagorean scale and "$\sqrt{2}$"



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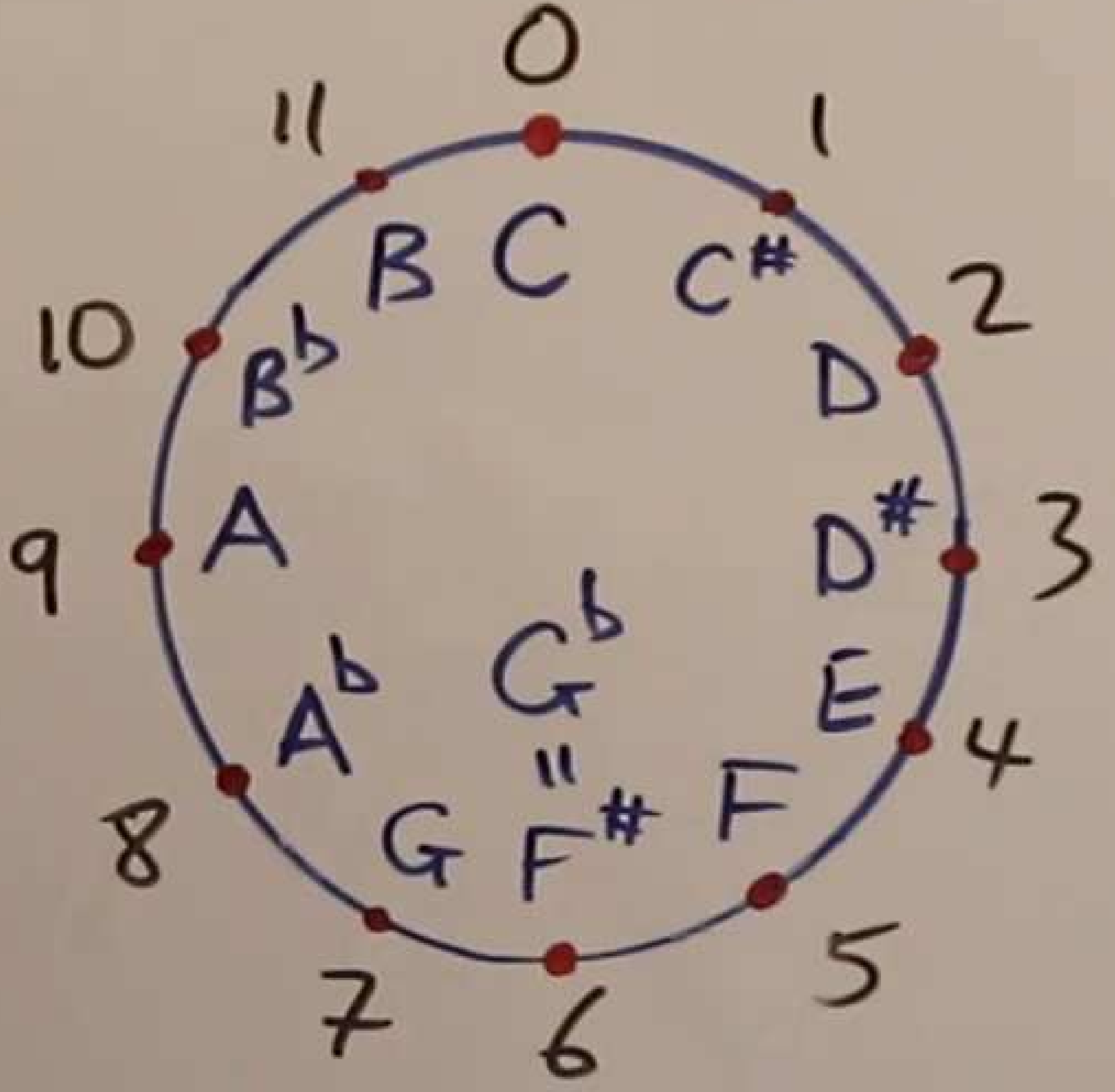
...e a 'perfect' Pythagorean

...ct fifths / 7-steps

...uency proportions?

The impossibilities of a Pythagorean scale and "$\sqrt{2}$"



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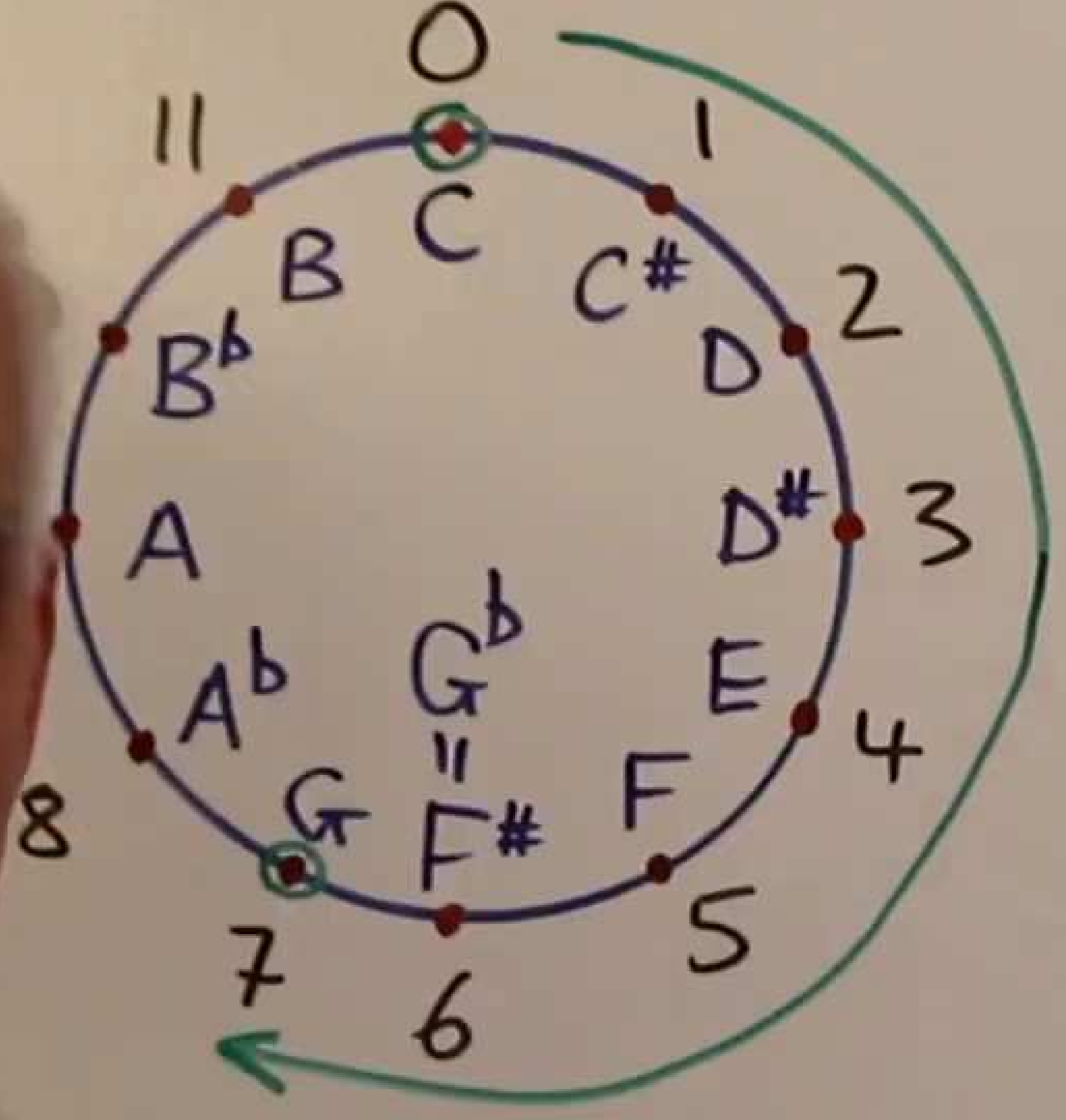
create a 'perfect' Pythagorean

perfect fifths / 7-steps

frequency proportions?

Cycle of fifths (up)



of 7-steps.

M note	note	formula freq.	eval. freq.	approx. freq.
0	C	1	1	1
7	G	$\frac{3}{2}$	$\frac{3}{2}$	1.5
14=2	D	$\frac{1}{2}\left(\frac{3}{2}\right)^2$	$\frac{9}{8}$	1.125
21=9	A	$\frac{1}{2}\left(\frac{3}{2}\right)^3$	$\frac{27}{16}$	~1.6875
28=4	E	$\frac{1}{4}\left(\frac{3}{2}\right)^4$	$\frac{81}{64}$	~1.2656
35=11	B	$\frac{1}{4}\left(\frac{3}{2}\right)^5$	$\frac{243}{128}$	~1.8984
42=6	F#	$\frac{1}{8}\left(\frac{3}{2}\right)^6$	$\frac{729}{512}$	~1.4238

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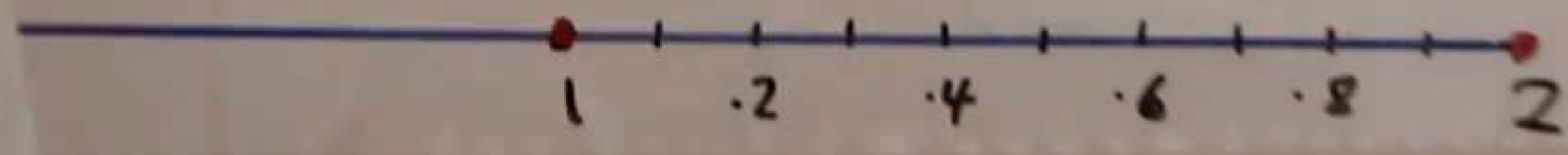
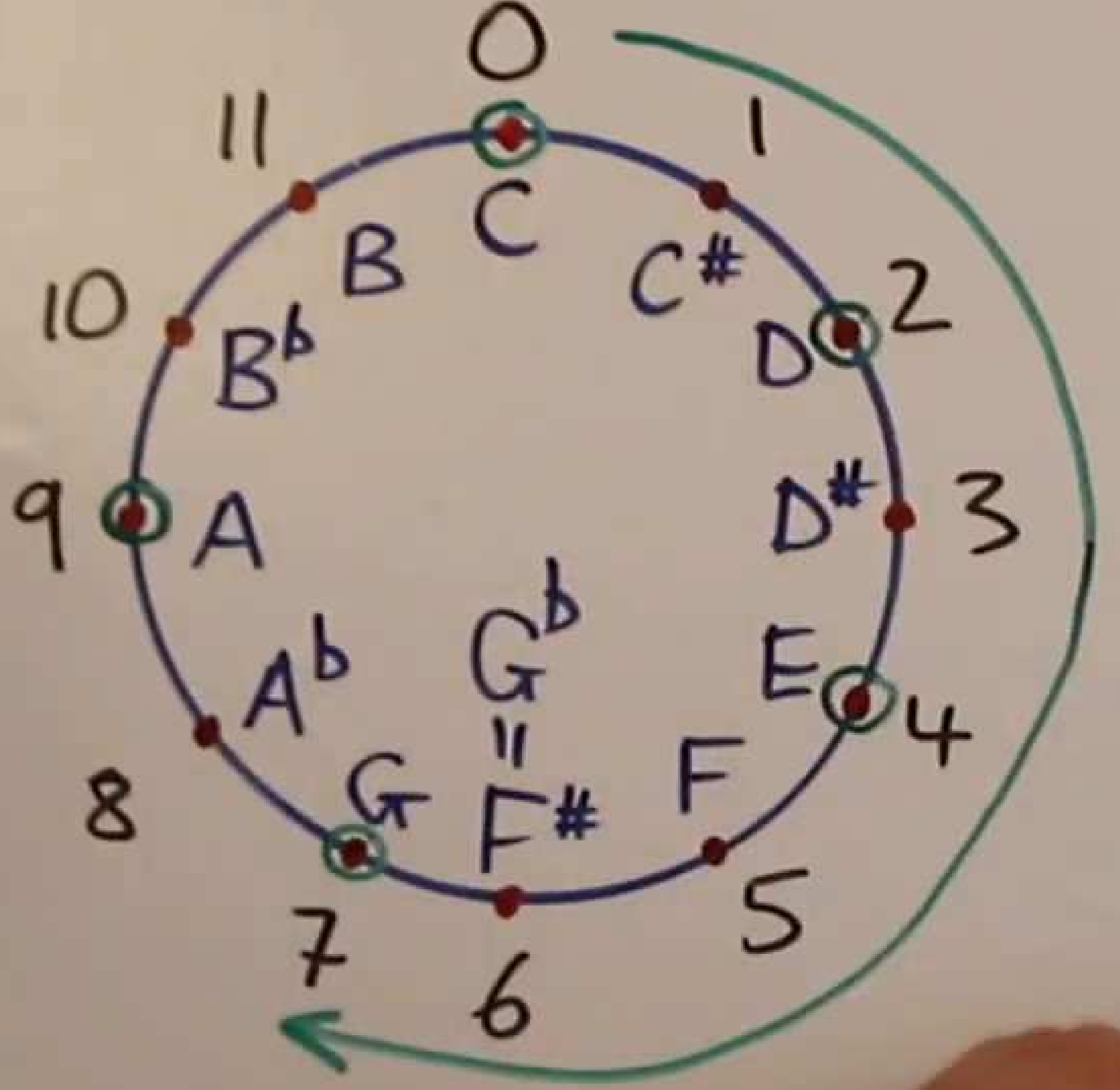


(freq. ⇒)

Cycle of fifths (up)



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[illegible]2 = 6	F#	$\frac{1}{8}\left(\frac{3}{2}\right)^6$	$\frac{729}{512}$	~1.4238

Cycle of 7

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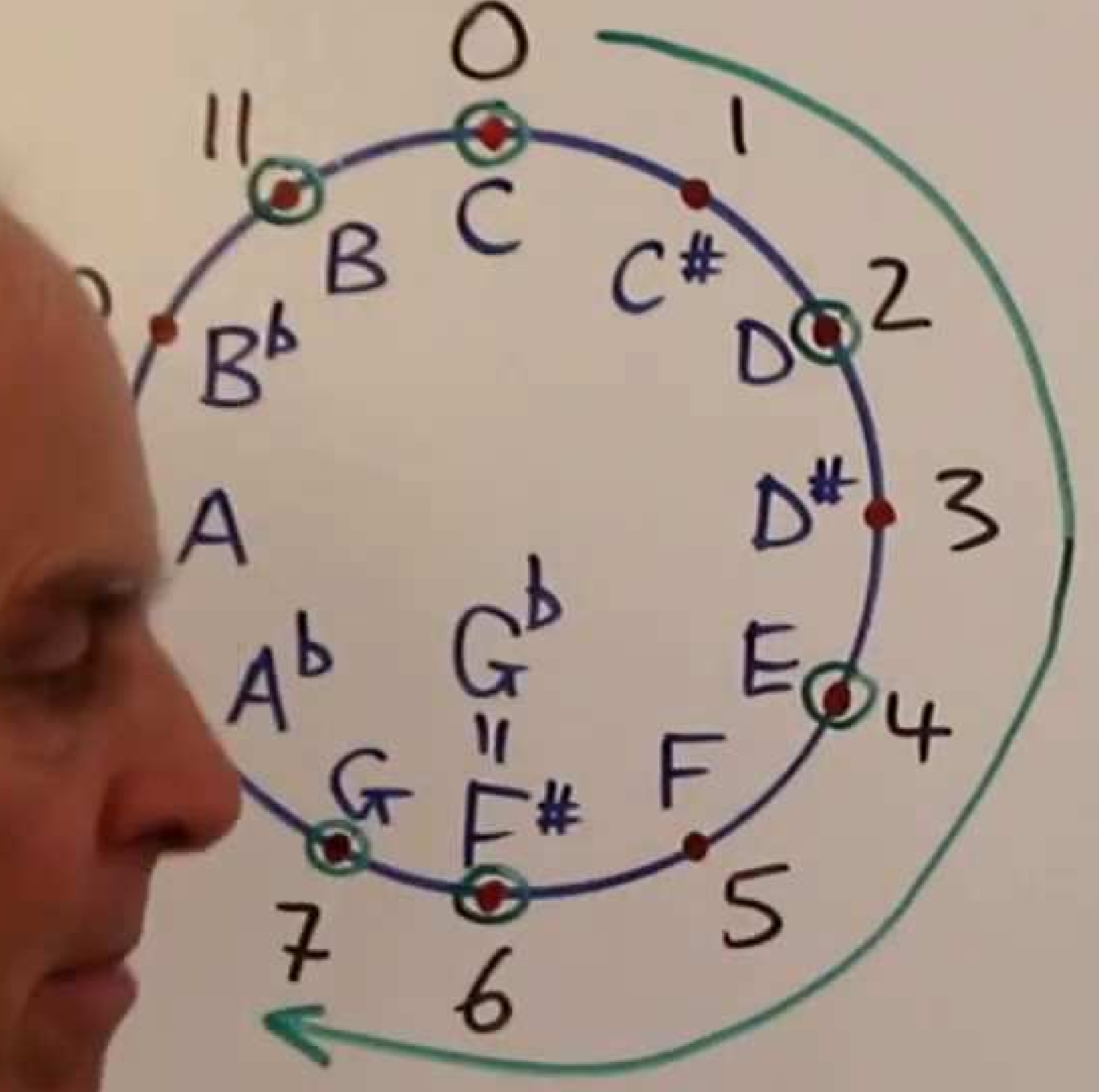
(freq. ⇒)

Cycle of fifths (up)



cle of 7-steps

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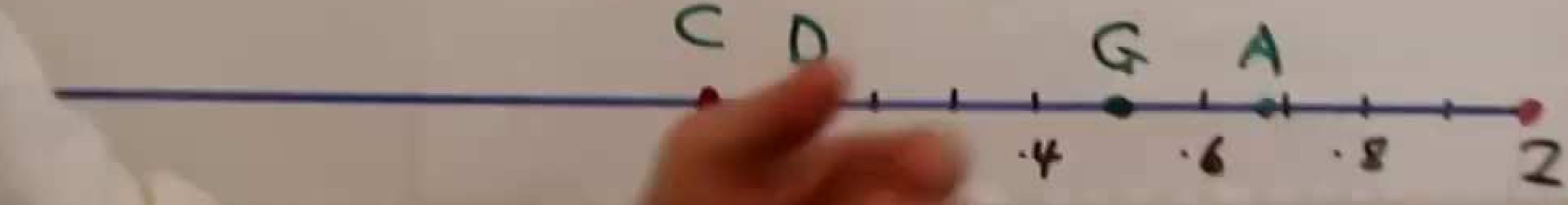

(freq. ⇒)

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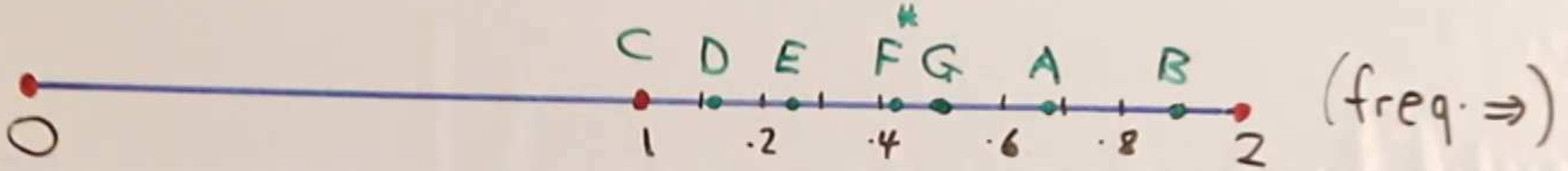

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Cycle of fifths (down)

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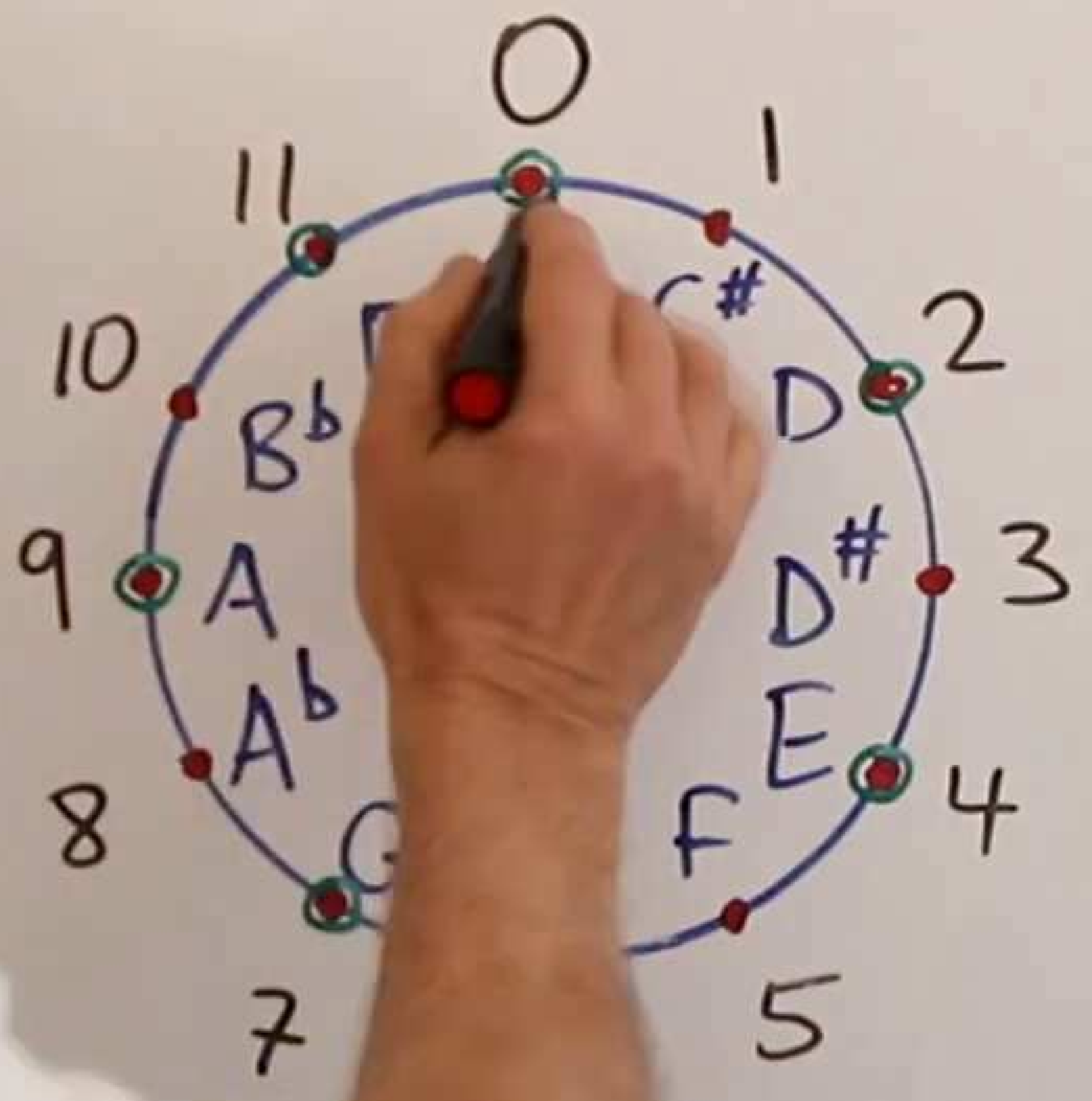
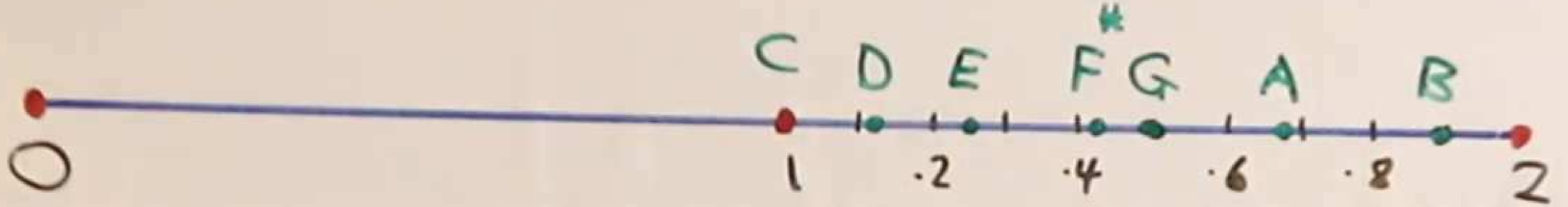
M



7-steps

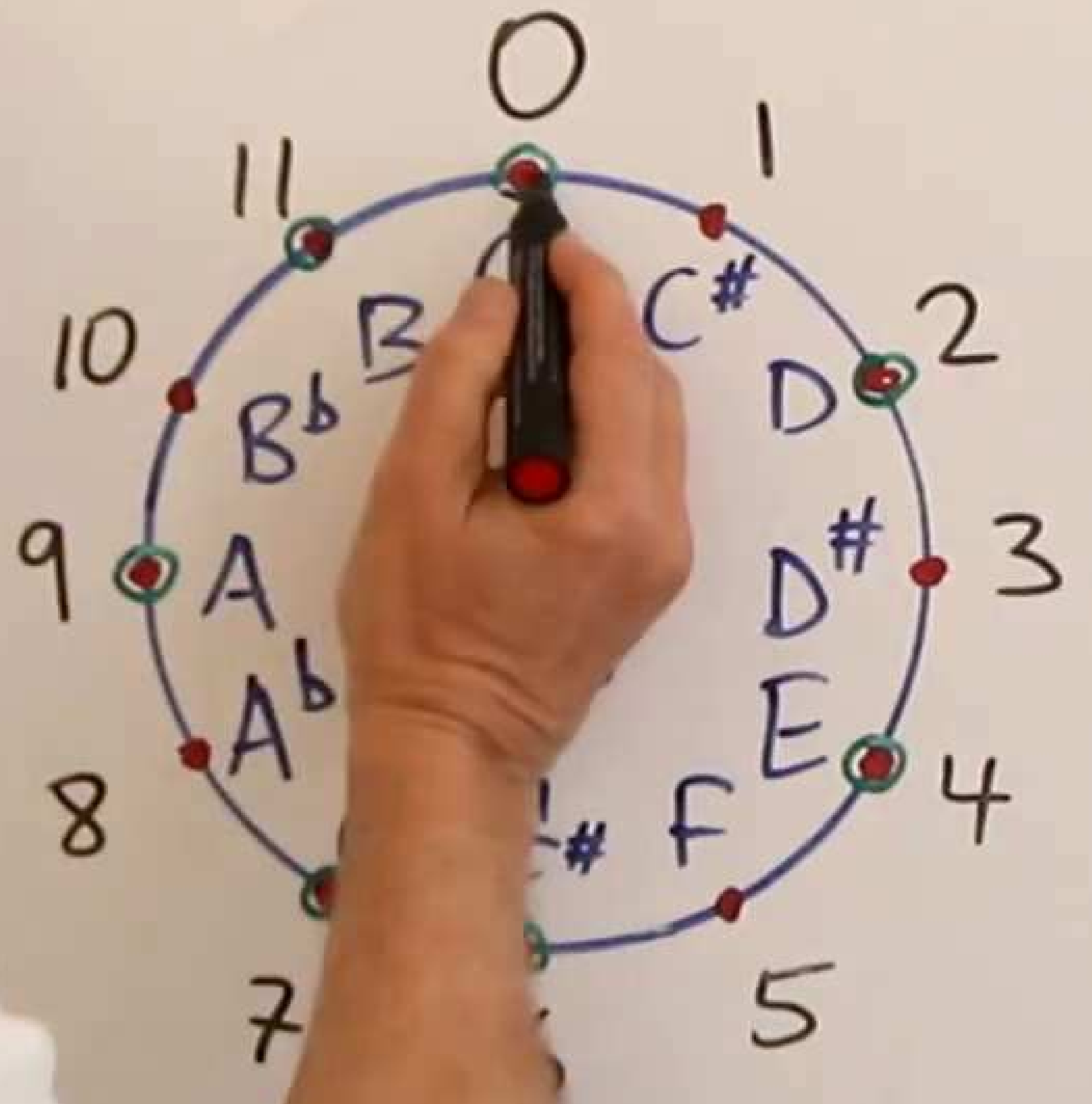
note	note			
0	C	1	1	1
$-7=5$	F	$2\left(\frac{2}{3}\right)$	$\frac{4}{3}$	~ 1.333
$-14=10$	B^b	$4\left(\frac{2}{3}\right)^2$	$\frac{16}{9}$	~ 1.7778
$-21=3$	E^b	$4\left(\frac{2}{3}\right)^3$	$\frac{32}{27}$	~ 1.1852
$-28=8$	A^b	$8\left(\frac{2}{3}\right)^4$	$\frac{128}{81}$	~ 1.5802
$-35=1$	D^b	$8\left(\frac{2}{3}\right)^5$	$\frac{256}{243}$	~ 1.0535
$-42=6$	G^b	$16\left(\frac{2}{3}\right)^6$	$\frac{1024}{729}$	~ 1.4047



(freq ⇒) ②

Cycle of fifths (down) ③

M



7-steps

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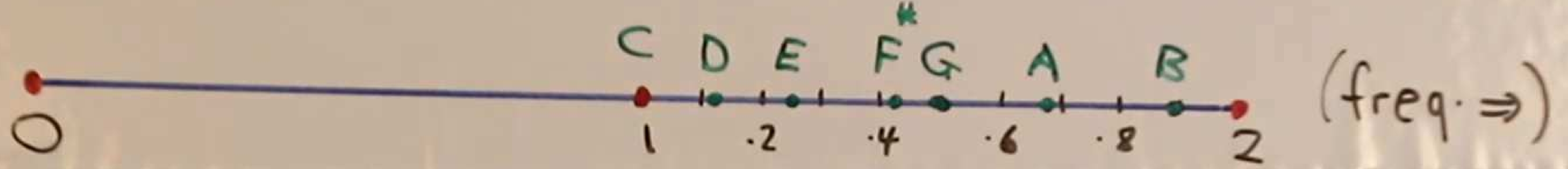

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Cycle of fifths (down)

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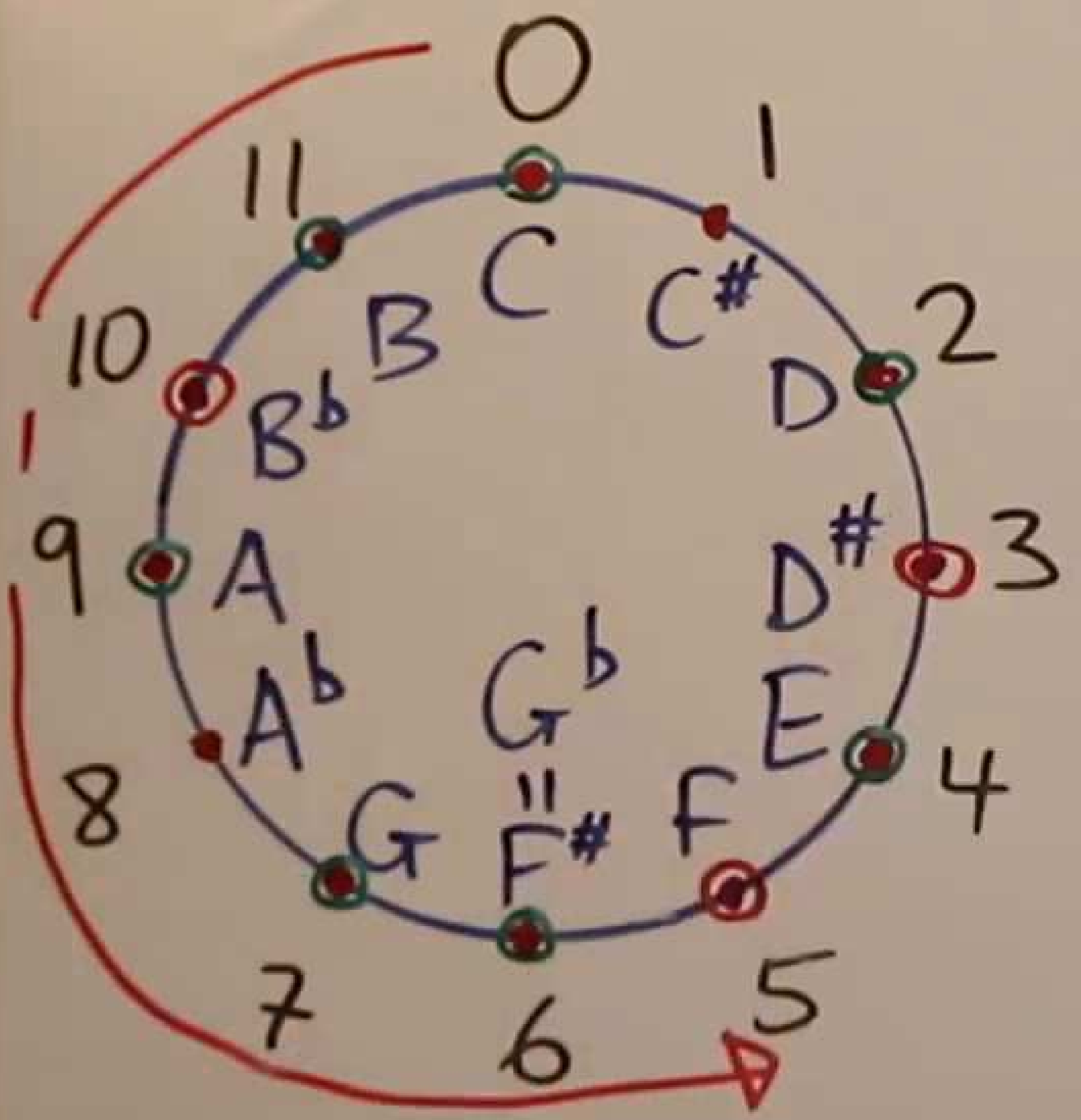
M



Cycle of 7-steps

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-14 = 10	B^b	$4\left(\frac{2}{3}\right)^2$	$\frac{16}{9}$	~1.7778
-21 = 3	E^b	$4\left(\frac{2}{3}\right)^3$	$\frac{32}{27}$	~1.1852
-28 = 8	A^b	$8\left(\frac{2}{3}\right)^4$	$\frac{128}{81}$	~1.5802
-35 = 1	D^b	$8\left(\frac{2}{3}\right)^5$	$\frac{256}{243}$	~1.0535
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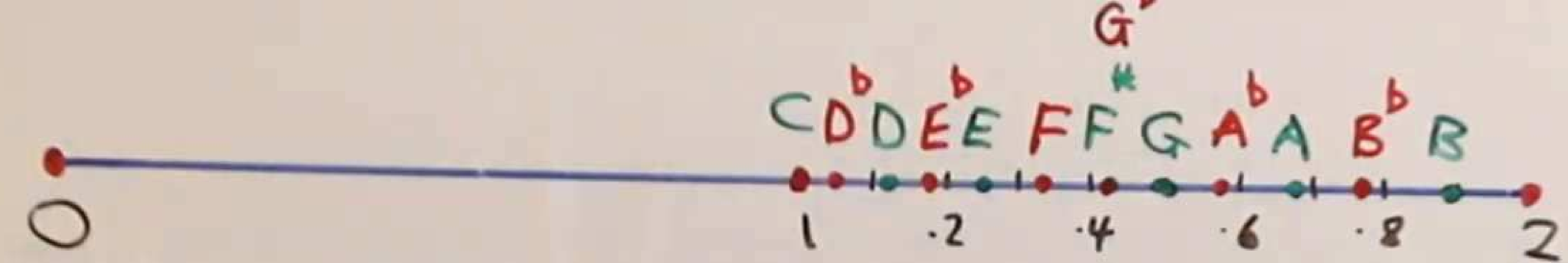

(freq. ⇒)

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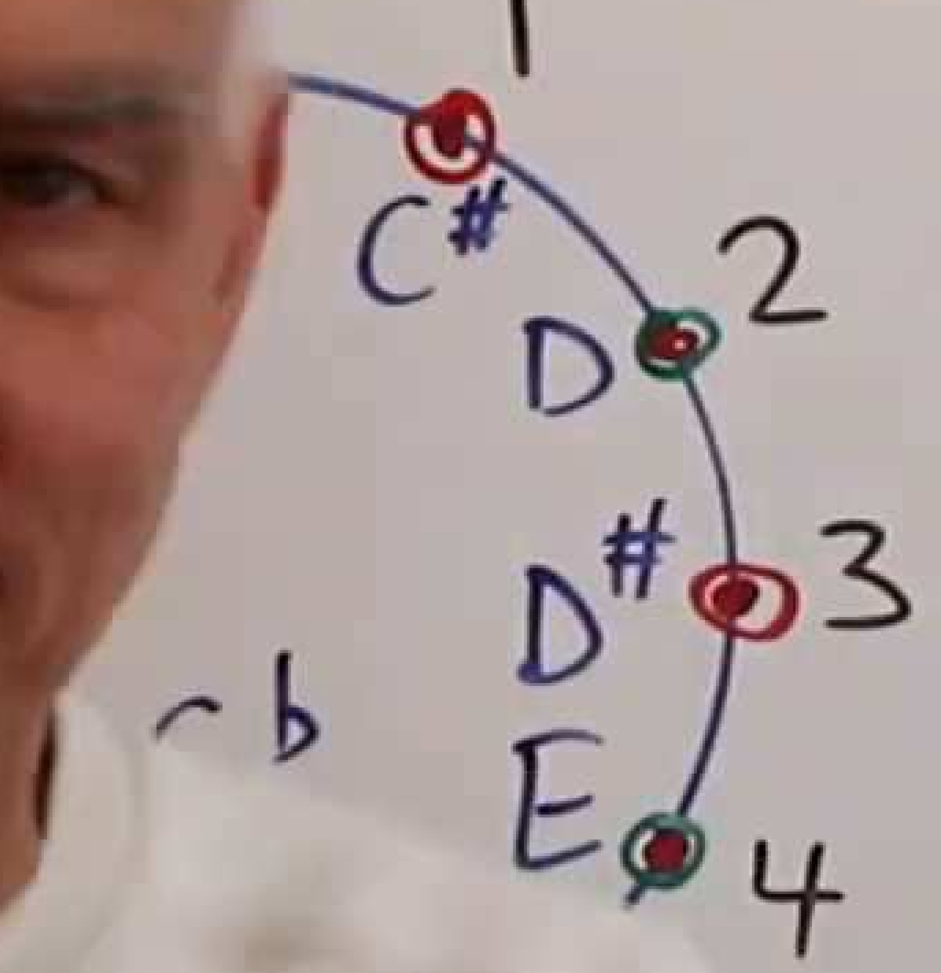

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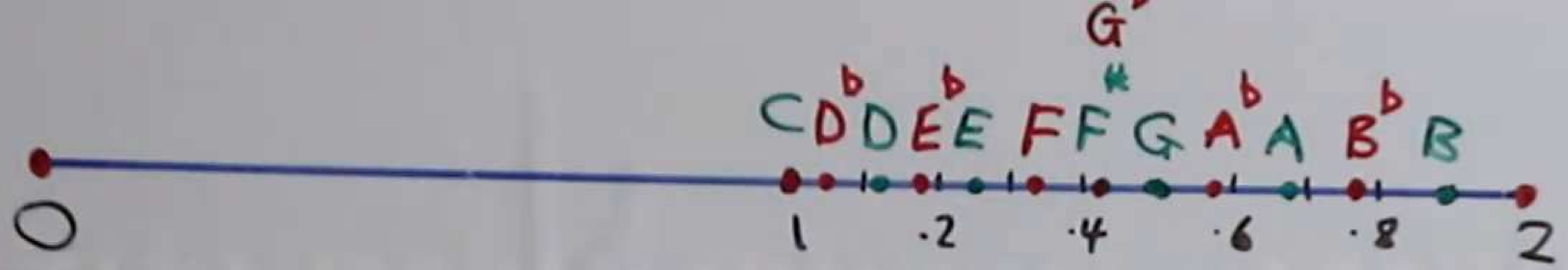

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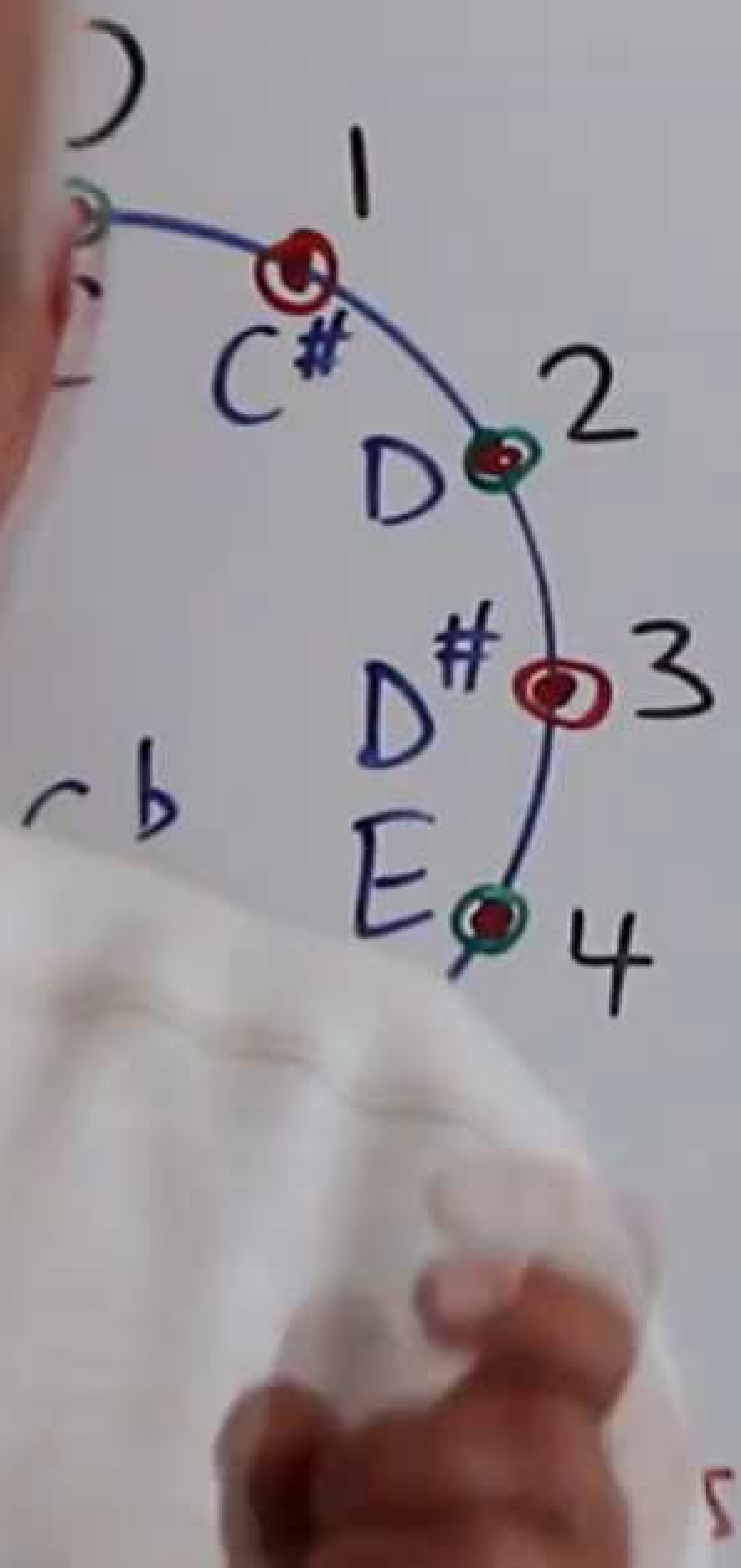

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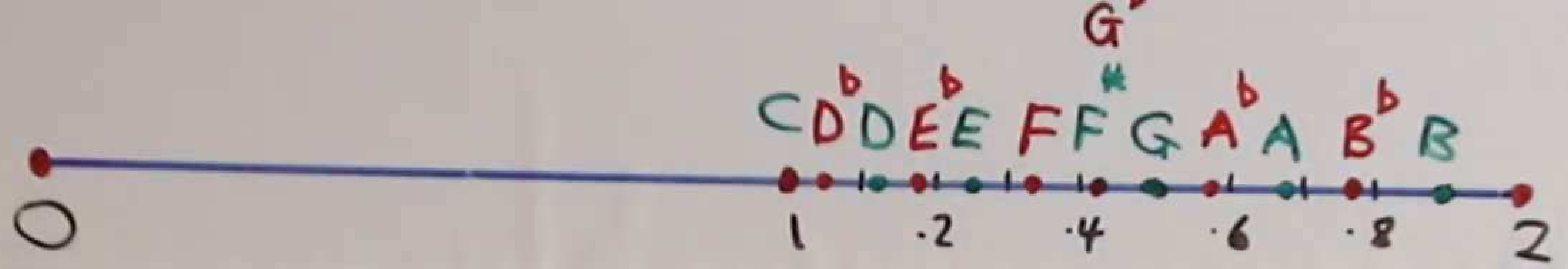

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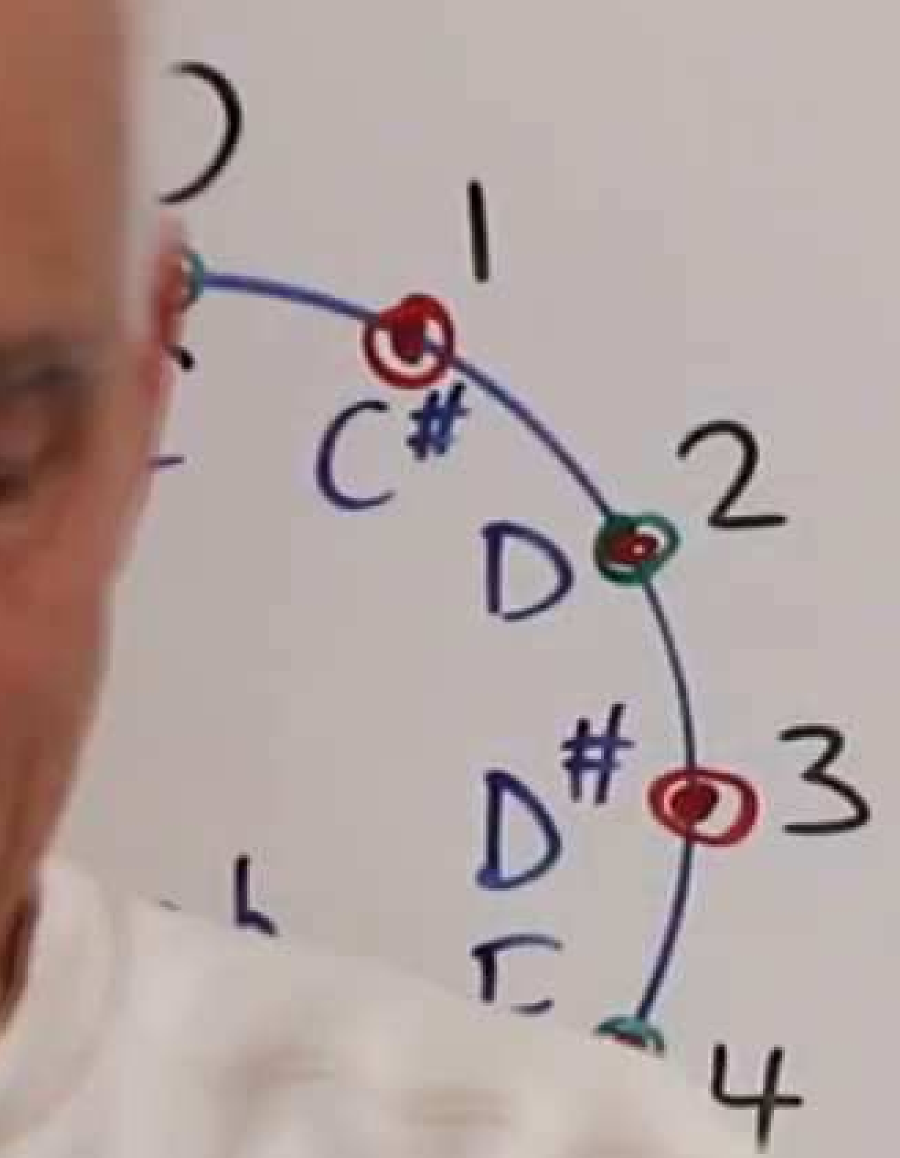

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